

ACKNOWLEDGEMENT

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ABSTRACT

Online fashion shopping has become very popular, but product returns are an important challenge for online businesses. Customers may return products because of size mismatch, fitting problems, product expectations, delivery issues and other factors. This project analyses a fashion order dataset containing 50,000 records and 25 attributes. The target variable is returned, where 1 represents a returned order and 0 represents an order that was not returned. The dataset has an observed return rate of 28.13%. Python is used for data analysis and visualization. After understanding and preprocessing the dataset, exploratory data analysis is performed to study return behaviour. Feature engineering is then used to create useful information such as effective price. Finally, Logistic Regression is used as a binary classification model to predict whether an order is likely to be returned. The main aim of this project is to understand return patterns and demonstrate how machine learning can be used for return prediction.

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CHAPTER 1: INTRODUCTION

Fashion e-commerce generates a large number of orders every day. Some of these orders are returned by customers due to different reasons such as incorrect size, unsuitable fitting, product expectations and delivery-related issues.

Analysing return behaviour can help businesses understand customer purchasing patterns and improve their services.

In this project, a fashion order dataset from Kaggle is analysed using Python. Different data analysis and visualization techniques are used to understand the dataset. A Logistic Regression model is then applied to predict the return status of an order.

1.1 PROBLEM STATEMENT

The main problem addressed in this project is:

To analyse fashion order data and predict whether an order will be returned or not using Logistic Regression.

The project studies important factors such as product category, price, discount, size mismatch, customer history, ratings and delivery time.

CHAPTER 2. OBJECTIVES

The main objectives of the project are:

- To understand the structure of the fashion returns dataset.
- To inspect the different attributes present in the dataset.
- To perform data cleaning and preprocessing.
- To analyse return behaviour using graphs.
- To study important factors related to returned orders.
- To create useful derived features from existing data.
- To apply Logistic Regression for binary classification.
- To evaluate the model using suitable performance measures.
- To understand the results obtained from the analysis.
- To identify possible applications of return prediction.

2.1 SCOPE

The project can be useful for:

- Understanding customer return behaviour.
- Studying size and fitting problems.
- Analysing product-related return patterns.
- Supporting better size recommendations.
- Understanding possible return risk.
- Improving business planning.

CHAPTER 3: DATASET DESCRIPTION

The dataset used in this project is:

Will This Order Come Back?

Source: **Kaggle**

The dataset contains:

- **50,000 records**
- **25 columns**
- Order-level information
- Product information
- Customer information
- Delivery information
- Return information

Each row represents an individual order record. The target variable is **returned**.

3.1 Important Attributes

Attribute	Description
order_id	Unique order identifier
category	Product category
fabric	Product material
price_usd	Product price
discount_pct	Discount percentage
size_ordered	Size selected
size_usual	Customer's usual size
size_mismatch	Indicates size mismatch
ordered_multiple_sizes	Multiple-size ordering indicator

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fit_type	Product fitting type
customer_prior_orders	Previous orders
customer_prior_return_rate	Previous return rate
avg_review_rating	Average product rating
days_to_delivery	Delivery time
return_reason	Reason for return
returned	Target variable

[Table 3.1: important fields from dataset]

CHAPTER 4: TOOLS AND TECHNOLOGIES

4.1 Software

- Python
- Jupyter Notebook / Kaggle Notebook
- Microsoft Word

4.2 Python Libraries Used

No.	Library	Purpose
1	Pandas	Data loading and data manipulation
2	NumPy	Numerical calculations
3	Matplotlib	Data visualization
4	Seaborn	Statistical visualization
5	Scikit-learn	Machine learning and model evaluation
6	SHAP	Model explainability
7	LIME	Local model explanation

[Table 4.1: List of libraries used]

CHAPTER 5: METHODOLOGY

The project follows these steps:

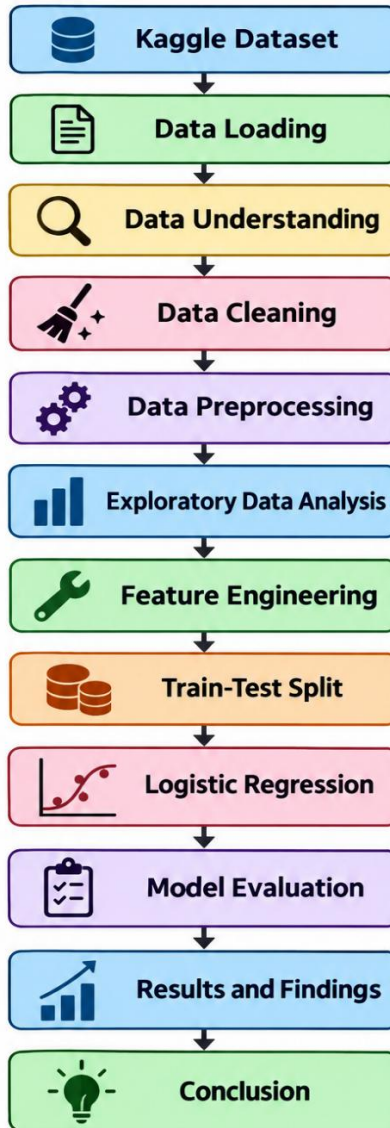


Figure 1: Project methodology workflow

CHAPTER 6: DATA UNDERSTANDING AND PREPROCESSING

The first step is to understand the dataset before applying machine learning.

The dataset is loaded and inspected to understand:

- Number of rows and columns
- Column names
- Data types
- Numerical and categorical variables
- Missing values
- Duplicate records
- Target variable

The target variable **returned** is separated from the input features for machine learning.

6.1 Data Preprocessing

The following preprocessing activities are performed:

1. Checking the dataset structure.
2. Checking missing values.
3. Checking duplicate records.
4. Checking data types.
5. Preparing categorical variables.
6. Preparing numerical variables.
7. Separating input features and target.
8. Dividing the dataset into training and testing data.

CHAPTER 7: EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis (EDA) is performed to understand the dataset using graphs and statistical information.

EDA helps identify patterns and differences in customer orders and return behaviour.

7.1 Category and Return Analysis

The category-wise graph shows the number of orders in different product categories and compares returned and non-returned orders.

This helps understand how return records are distributed across product categories.

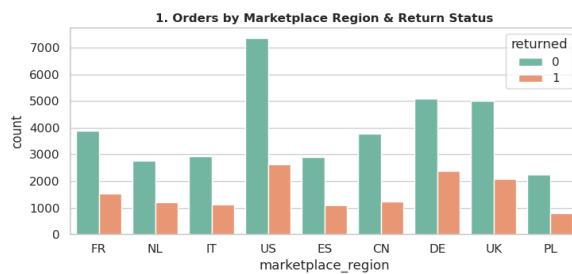


Figure 7.1: count v/s marketplace region



Figure 7.2: count v/s category

7.2 Numerical Feature Analysis

Numerical variables help us understand the distribution and differences between returned and non-returned orders.

Important variables include:

- Price
- Previous return rate
- Delivery time
- Review rating
- Previous orders
- Size mismatch

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The graphs help identify the spread, median and possible extreme values.

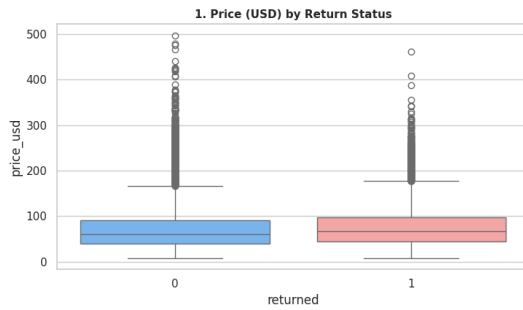


Figure 7.3: price_usd v/s returned

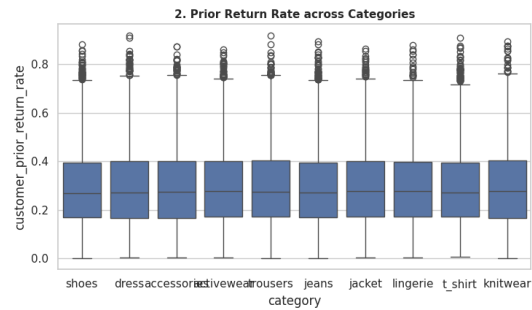


Figure 7.4: return rate v/s category

7.3 Return Behaviour Comparison

The next graph compares numerical variables for different return statuses. The visualization helps understand how numerical variables are distributed among returned and non-returned orders.

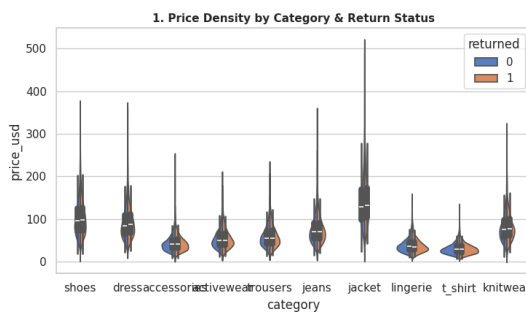


Figure 7.5: price_usd v/s category

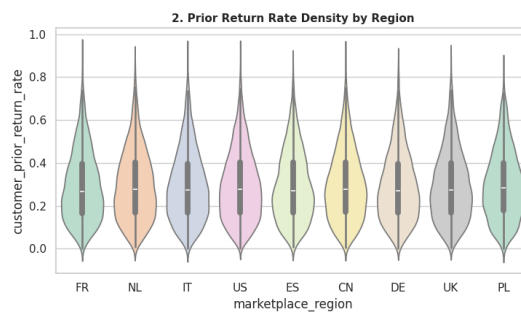


Figure 7.6: return rate v/s marketplace_region

7.4 Relationship Between Numerical Features

A pair plot is used to study relationships between important numerical variables.

The visualization compares variables such as:

- Price
- Customer prior return rate
- Delivery days
- Average review rating
- Size mismatch

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The returned status is used to distinguish different groups.

The plot helps identify possible relationships between variables and return behaviour.

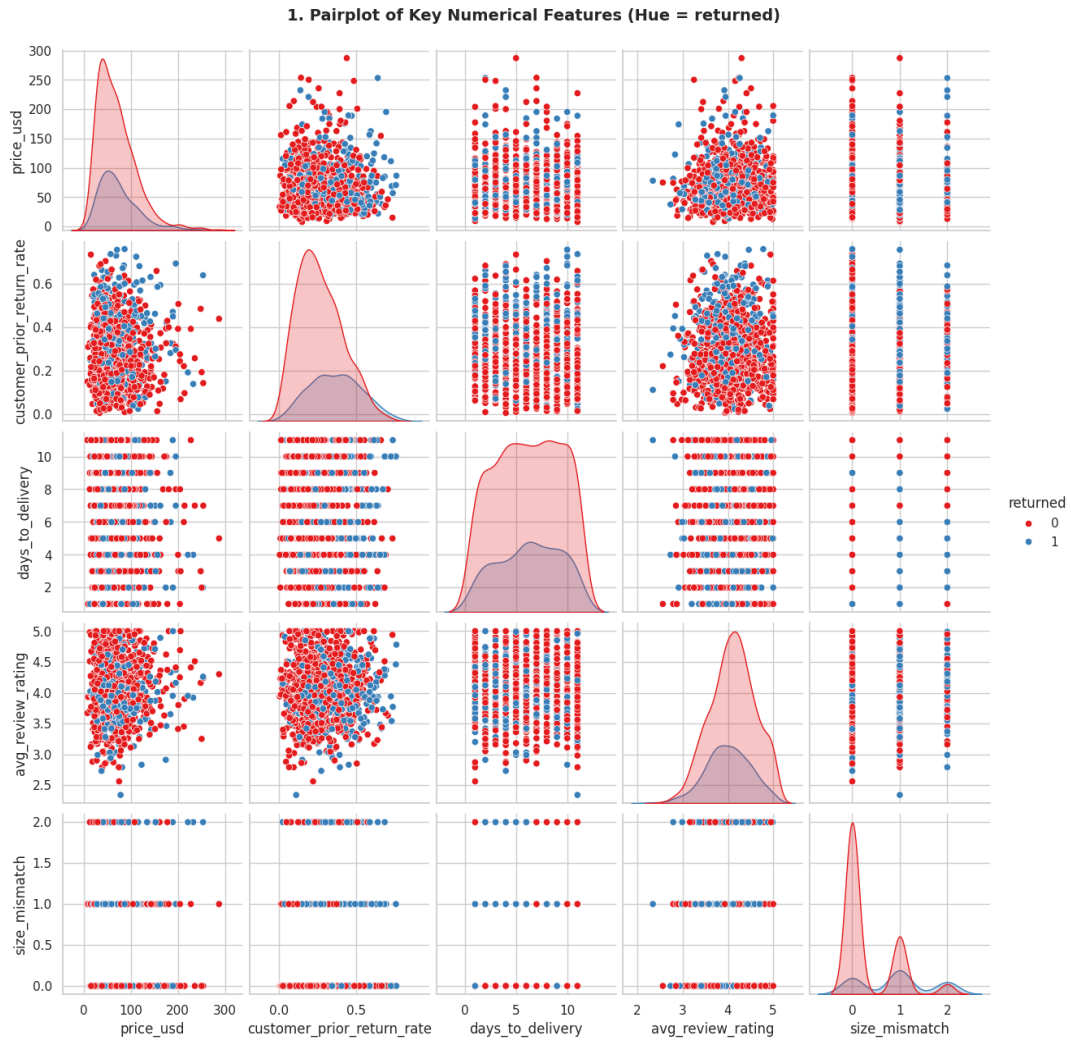


Figure 7.5: Pair Plot of Important Numerical Features by Return Status

CHAPTER 8: FEATURE ENGINEERING

Feature engineering means creating a new useful variable from existing variables.

In this project, **Effective Price** is created using the product price and discount.

8.1 Effective Price

$$\text{Effective Price} = \text{price_usd} \times (1 - \text{discount_pct} / 100)$$

Effective price represents the estimated price after applying the discount.

It provides an additional variable for analysing the relationship between product price and return behaviour.

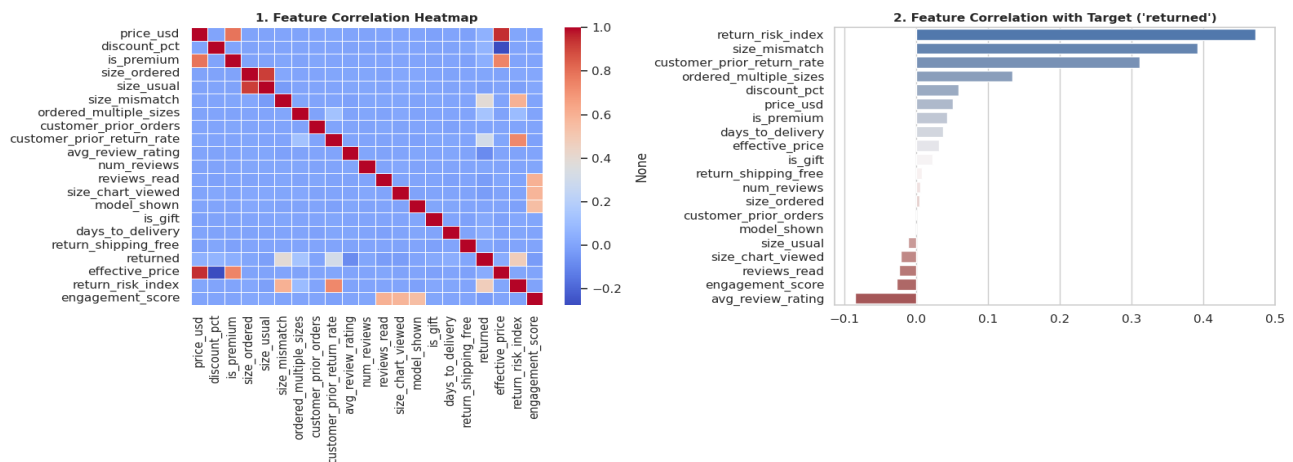


Figure 8.1: Feature Engineering and Feature Relationship Analysis

CHAPTER 9: LOGISTIC REGRESSION MODEL

Logistic Regression is a supervised machine learning algorithm used for classification problems. In this project, it is used to predict the binary target variable:

- **0 → Order not returned**
- **1 → Order returned**

The prepared dataset is divided into training and testing data.

The training data is used to build the Logistic Regression model, while the testing data is used to evaluate how well the model performs on unseen records.

9.1 Why Logistic Regression?

Logistic Regression is suitable for this project because:

- The target variable has two classes.
- It is simple to understand.
- It is suitable for binary classification.
- It provides prediction probabilities.
- It is commonly used as a classification baseline.

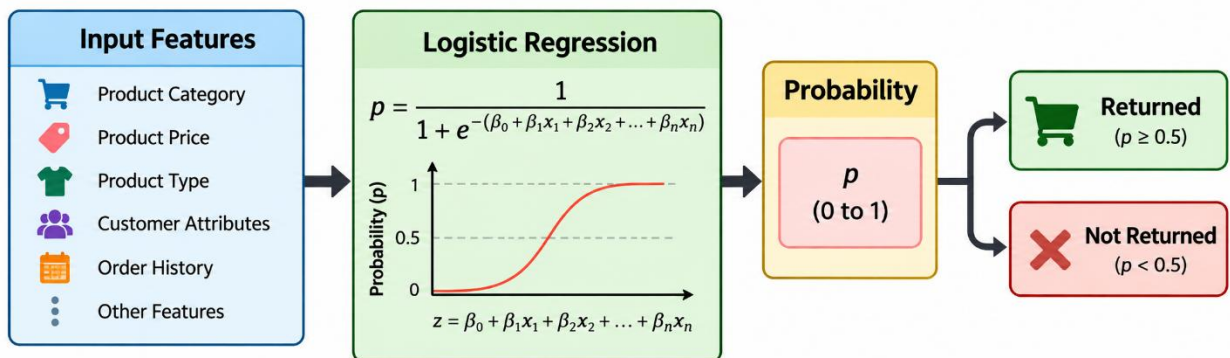


Figure 9.1: Logistic Regression concept diagram.

CHAPTER 10: MACHINE LEARNING APPROACH

The project uses classification models to predict the binary target variable “returned”. The notebook includes different models so that their performance can be compared.

10.1 Models

- Logistic Regression – used as a simple baseline classification model.
- Random Forest – combines multiple decision trees and can model nonlinear relationships.
- Gradient Boosting – builds trees sequentially to improve prediction.
- Ensemble models – methods such as AdaBoost, Bagging and Extra Trees are used for comparison.

10.2 Evaluation Measures

- Accuracy – proportion of correctly classified records.
- Precision – proportion of predicted returns that are actually returns.
- Recall – proportion of actual returns identified by the model.
- F1-score – combined measure of precision and recall.
- ROC-AUC – measures ranking performance across classification thresholds.

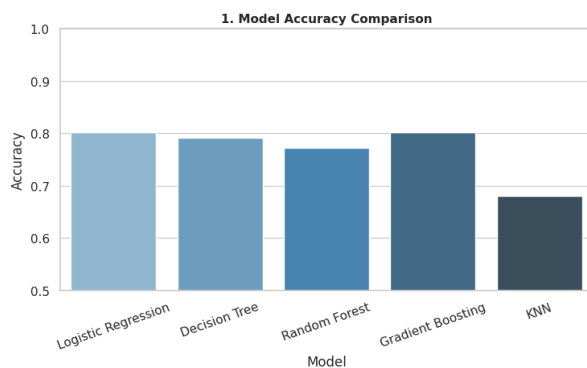


Figure 10.1: model accuracy comparison

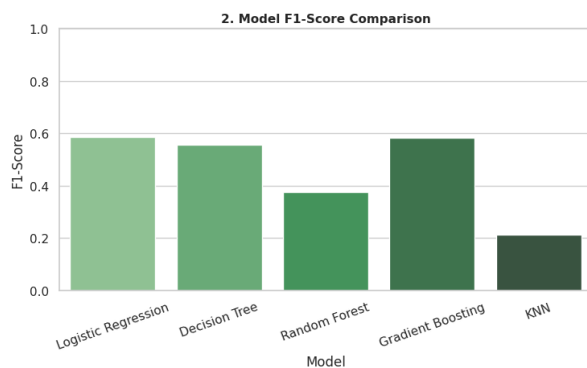


Figure 10.2: F1 score comparison

10.3 interpretation

The model evaluation results show the classification performance of the different machine learning models. Accuracy, F1-score, ROC-AUC and precision-recall curves are used to compare the models. The confusion matrix provides information about correctly and incorrectly classified returned and non-returned orders. These measures provide a broader understanding of model performance than accuracy alone.

CHAPTER 11: MODEL INTERPRETATION AND ADVANCED ANALYSIS

Model interpretation is used to understand which variables contribute to the predictions. The notebook includes feature importance and explainability methods such as SHAP and LIME. These methods explain model behaviour but should not be considered proof of causation.

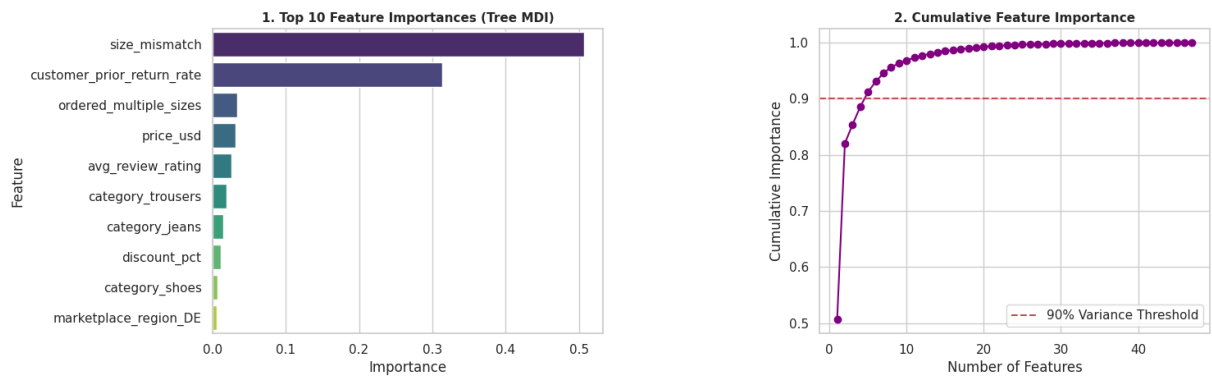


Figure 11.1: Feature importance and explainability

11.1 Anomaly Detection

Anomaly detection identifies records that are different from common patterns. The notebook includes techniques such as Isolation Forest and Local Outlier Factor. An unusual record is not necessarily an error; it should be reviewed using domain knowledge.

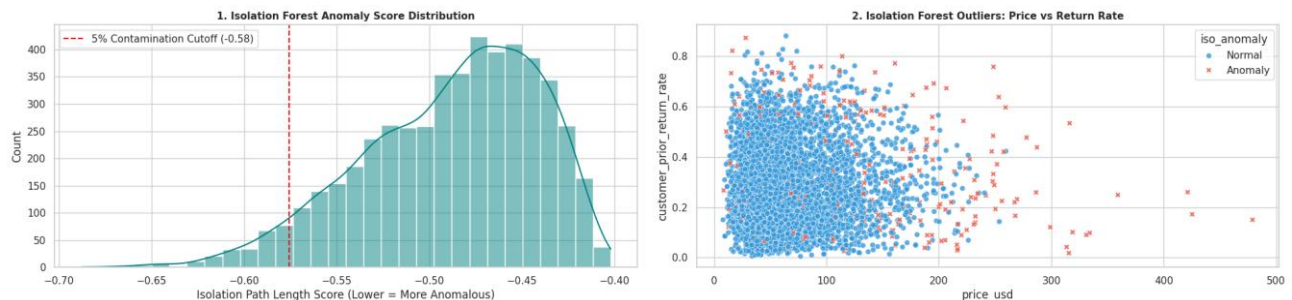


Figure 11.2: Anomaly detection visualisation

11.2 CLUSTERING

Clustering groups records based on similarity without directly using the target label. The notebook uses clustering methods such as K-Means, DBSCAN and Agglomerative Clustering. Clustering may help identify groups of customers or orders with similar characteristics.

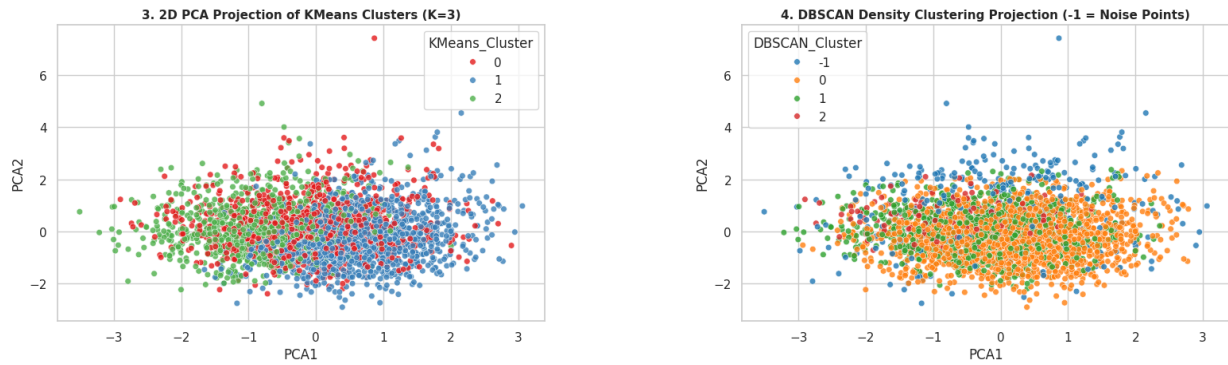


Figure 11.3: Clustering visualisation

11.3 DIMENSIONALITY REDUCTION

Dimensionality reduction methods such as PCA and t-SNE are used to represent complex data in fewer dimensions. These visualisations make it easier to inspect patterns and groups, although the reduced view does not contain every detail of the original dataset.

CHAPTER 12: RESULTS AND DISCUSSION

- The dataset contains **50,000 records and 25 attributes**.
- The overall return rate is **28.13%**.
- The average product price is **\$72.22** and the median price is **\$61.78**.
- The average discount is **15.49%**.
- The average delivery time is **6.01 days**.
- Size-related features provide useful information for analysing possible fit problems.
- Customer history can be considered while studying repeated return behaviour.
- Different machine learning models allow comparison of classification performance.
- Explainability and clustering provide additional understanding beyond a single prediction value.

CHAPTER 13: APPLICATIONS AND FUTURE SCOPE

13.1 Applications

- Improved size recommendations before checkout.
- Identification of orders that may require additional fit guidance.
- Product and fabric analysis for reducing repeated returns.
- Better product descriptions and size charts.
- Planning of reverse-logistics and customer-service workload.

13.2 Future Scope

- Use time-based train-test validation.
- Apply natural language processing to review text and return reasons.
- Create a dashboard for category, region and fit-based filtering.
- Deploy the model through an API after proper validation.
- Monitor model drift and fairness across customer groups.
- Compare predicted return risk with actual future returns.

13.3 LIMITATIONS

- The dataset is observational, so relationships should not automatically be considered causal.
- The dataset may not include every reason for a return.
- Model performance may change for another marketplace or time period.
- Time-based validation and monitoring would be needed for real deployment.
- Business decisions should be tested before applying model predictions to customers.

CONCLUSION

This project demonstrates the use of Python for analysing fashion order returns and applying machine learning for prediction. The dataset contains 50,000 records and 25 attributes, with an observed return rate of 28.13%. The project includes data understanding, preprocessing, exploratory data analysis, feature engineering, and Logistic Regression. The analysis helps us understand how product, customer, and order-related information can be used to study return behaviour. Logistic Regression provides a simple and understandable approach for predicting whether an order is returned or not. Overall, the project helped us understand the complete basic workflow of a Data Science project using Python.

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